

Emmanuel Idoko

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Research interests: representation learning & self-supervised methods · interpretability · robustness and trustworthy ML

Education

University of Lagos 2021 – 2027
B.Sc. Computer Science · CGPA: 4.68 / 5.0 Lagos, Nigeria

- Relevant coursework: Linear Algebra, Calculus, Discrete Mathematics, Data Structures, Algorithms & Complexity, Operating Systems, Concurrent Programming, Operations Research.

African Computer Vision Summer School (ACVSS) — 3rd Edition Jul 2026
Selected Participant · Google AI Community Center Accra, Ghana

- Competitively selected; curriculum spanned self-supervised learning, generative modelling, multimodal and 3D vision, and fairness & ethics, with keynotes from researchers at Oxford, EPFL, Amsterdam, Michigan, and Google DeepMind.
- Presented SharpXR at the poster session and competed in the research hackathon, building a task-aware super-resolution pipeline for satellite flood mapping (project below).

Udacity Oct 2022 – Jan 2024
AI Programming with Python Nanodegree Remote

Research & Publications

Retrieval-Induced Hallucination in Medical Vision-Language Models Submitted — NeurIPS 2026
First author · Africa in AI Workshop (Springer LNCS)

- Found 95% of retrieval-augmented chest X-ray reports copy text verbatim from another patient's report (0% without retrieval); traced the cause to embedding similarity tracking anatomy, not disease.
- Showed retrieval doubles clinical accuracy when relevant (CheXbert F1 0.201→0.402) and collapses it to 0.043 when mismatched, over 100 MIMIC-CXR studies.
- Built the full pipeline (BioMedCLIP retrieval, LLaVA-1.5-7B, CheXbert evaluation) with patient-level leakage controls and a coincidental-overlap control ruling out chance agreement as the source of the copying result; bit-for-bit reproducible.

VAMAE: Vessel-Aware Masked Autoencoders for OCT-Angiography Accepted — ICPR 2026
arXiv:2604.06583

- Proposed a self-supervised masked-autoencoder (MAE) variant with a vessel-aware reconstruction loss that prioritizes fine vascular structures, learning robust retinal representations for downstream segmentation and classification.

SharpXR: Structure-Aware Denoising for Pediatric Chest X-Rays Accepted — MICCAI 2025 (MIRASOL)
arXiv:2508.08518

- Benchmarked seven denoising baselines (RED CNN, DnCNN, HFormer, ResUNet++, Attention U-Net, Sharp U-Net, BM3D); structure-preserving denoising raised downstream pneumonia-classification accuracy from 88.8% to 92.5%.
- Led the abstract and introduction and edited the full manuscript; presented as a poster at MIRG-ICAIR 2025 and the ACVSS 2026 poster session.

Research Experience

ML Collective — Independent ML Researcher Sep 2024 – Present
Remote

- Present and critically analyze seminal and contemporary ML papers weekly — reproducing methods and interrogating experimental design — with a focus on medical imaging and clinical NLP.
- Collaborate in focus groups to formulate hypotheses and design experimental protocols for computer-vision research projects.

AidCare — Founder & AI/ML Engineer May 2025 – Present
Lagos, Nigeria (Remote)

- Architected a dual-mode Retrieval-Augmented Generation (RAG) system for clinical decision support, with separate knowledge bases for physicians and community health workers.
- Built a semantic-search pipeline (Sentence Transformers + FAISS) indexing 500+ Nigerian clinical protocols for sub-second retrieval, and a token-efficient Gemini prompting layer for grounded differential-diagnosis generation.
- Integrated Whisper ASR and Tesseract OCR into a multimodal ingestion pipeline; deployed a containerized FastAPI backend on DigitalOcean with RBAC, scoped tokens, and anonymized logging.

Wema Bank — Data & AI Intern Sep 2024 – Dec 2024
Lagos, Nigeria (Hybrid)

- Built a governance-compliant pipeline extracting and analyzing 10,000+ Play Store reviews into structured product insights for cross-functional teams.
- Contributed to an AI-powered knowledge chatbot (Node.js, Azure) serving 2,000+ staff with instant access to internal documentation.

Engineering Experience

HabariPay – GTCO — Software Engineer

Apr 2025 – Present

Lagos, Nigeria (Hybrid)

- Build enterprise events software for Squad (GTCO's fintech arm), supporting event-management and payment workflows at scale.
- Shipped a full-stack vendor-guide app (Next.js, Node.js, PostgreSQL) used by 260+ attendees (2,100+ page views); implemented a TF-IDF recommendation engine with Gemini summaries, improving click-through by 7%.

NexaScale — Cloud Engineer Intern

Feb 2023 – May 2023

Remote

- Automated deployment of 30+ AWS resources with Terraform (~75% manual setup); built a Node.js microservice containerized with Docker and orchestrated on Kubernetes (AKS) via an Azure DevOps CI/CD pipeline.

Selected Projects

Task-Aware Super-Resolution for Flood Mapping — PyTorch, latent-diffusion SR, U-Net · ACVSS 2026 Hackathon (team of 3)

- Tested whether super-resolution actually improves downstream flood mapping from degraded satellite imagery, treating enhancement as an intermediate representation whose value is proven on a segmentation task, not a sharpness score; built a 3-phase pipeline (conditioned latent-diffusion SR → flood-aware fine-tuning → U-Net segmentation → per-tile risk ranking) on a SpaceNet-8 subset.
- Ran controlled experiments against degraded-LR and bicubic baselines on a shared task and split: diffusion SR improved every downstream metric (mean IoU +24%, flood mIoU +11%), while an earlier SR run hurt segmentation — isolating semantic fidelity vs. hallucinated structure (domain shift) as the condition under which enhancement helps.

Virtue Foundation Intelligence Platform — Python, LangGraph, FAISS, GPT-4o-mini, Next.js

- Built an end-to-end agentic platform analyzing 797 Ghana healthcare facilities across 16 regions; a LangGraph supervisor routes natural-language queries to 6 specialized sub-agents over a FAISS / OpenAI-embedding vector store.
- Designed an Intelligent Document Processing pipeline (GPT-4o-mini) with per-field confidence scoring; identified 10 medical deserts and flagged 43 anomalies. 2nd place — Databricks × Hack-Nation Global AI Hackathon.

Kinetix — Python, FastAPI, Next.js, Gemini API, WebSockets

- Built a real-time AI trainer using Gemini vision to classify posture and joint alignment frame-by-frame; designed a dual-persona agent architecture separating internal biomechanical reasoning from external coaching over low-latency WebSockets.

PheraCAM — Python, OpenCV, Azure, OpenAI API

- Real-time face-recognition system: >95% accuracy at <150 ms latency. 2nd place among 50+ teams — Hacklab Hackathon.

Technical Skills

Languages: Python, JavaScript / TypeScript, SQL, C, Bash

ML & Deep Learning: PyTorch, Hugging Face Transformers, Sentence Transformers, Scikit-learn, NumPy, FAISS

Methods: Vision Transformers, self-supervised learning (SimCLR, BYOL, DINOv2, MAE), video action recognition (VideoMAE, optical flow), retrieval-augmented generation, ML fairness auditing

Systems & Tools: FastAPI, Flask, Node.js, Next.js, Docker, Kubernetes, PostgreSQL, Git, Linux, AWS / Azure / DigitalOcean

Awards, Scholarships & Leadership

- Jim Ovia Undergraduate Scholarship (2025–2027) — national scholarship recognizing top STEM performance.
- PICFI Undergraduate Scholarship (2025–2027) — national STEM scholarship.
- Squad Hackademy Undergraduate Scholarship (2024–2027) — recognizing top programming performance.
- Project Lead, Unilag Data Community (2024–Present) — built and delivered a workshop series (Python, SQL, Power BI, Excel) training 250+ students.